Udhay Chowdhury

Aspiring Biosensor Researcher | Electrochemical & Wearable Biosensors | Sustainable Healthcare Innovator

Chattogram, Bangladesh

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Profile

Aspiring **wearable bioelectronics researcher** focused on developing self-powered, low-cost wearable sensors for early cardiometabolic disease detection in resource-constrained settings, leveraging strong skills in numerical device simulation, data-driven optimization, and embedded sensor systems.

Academic Credentials

B.Sc. in Electrical and Electronic Engineering Expected Graduation: June 2026

Chittagong University of Engineering & Technology (CUET), Chattogram, Bangladesh

CGPA: 3.66 / 4.00 (up to 6th semester)

Relevant Coursework

- Electronic Circuits; Microprocessor and Interfacing; Measurement and Instrumentation.
- Biomedical Instrumentation; Optoelectronics.
- Semiconductor Devices; Electromagnetic Fields and Waves; VLSI Design.

Research Experience

Undergraduate Thesis Research, CUET

- Designed and simulated Triboelectric Nanogenerator (TENG) structures integrating surface nanostructures and a charge-trapping layer for wearable self-powered applications.
- Optimized device geometry and material parameters to achieve a significant increase in simulated power density (around 300 mW/m^2, over 15% improvement compared to baseline designs).
- Analyzed charge distribution and interfacial fields using semiconductor and device-physics concepts to understand performance enhancement mechanisms.

Individual Research (STARLAB – IET, CUET)

Supervisor: Adnan Faisal Siddique, Faculty Member, IET, CUET

- Conducted machine-learning based prediction of optimal fabrication parameters for lead-free perovskite solar cells using curated experimental and simulated datasets.
- Performed data collection, cleaning, and implemented and compared Random Forest, XGBoost, and LightGBM models to improve power conversion efficiency prediction and guide process parameter selection.

Professional Experience

Industrial Training Trainee, Bangladesh Telecommunications Company Limited (BTCL)

2-week attachment, Chattogram, Bangladesh

- Observed digital switching systems, optical fiber network architecture (backbone and access), and GPON-based FTTH operation and maintenance, including fault detection and configuration of OLTs and ONUs.

Scholarly Work

Journal Article

- **Simulation and optimization of a $CsSnI_3/CsSnGeI_3/Cs_3Bi_2I_9$ based triple absorber layer perovskite solar cell using SCAPS-1D**.
  Umme Mabrura Umama, Mohammad Iftekher Ebne Jalal, Md Adnan Faisal Siddique, **Udhay Chowdhury**, Md Inzamam Ul Hoque, Md Jahidur Rahman.
  Journal of Physics and Chemistry of Solids, Vol. 198, March 2025, Article 112480.
  DOI: 10.1016/j.jpcs.2024.112480

Conference Presentations

- **Numerical modeling of Perovskite Tandem Solar Cell with $CH_3NH_3SnI_3$ Bottom Absorber**.
  Udhay Chowdhury, Mohammad Iftekher Ebne Jalal, Md Adnan Faisal Siddique, Umme Mabrura Umama, Md Inzamam Ul Hoque, Md Jahidur Rahman.
  4th International Conference on Electrical, Computer and Communication Engineering (ECCE 2025), CUET, Bangladesh.
  DOI: 10.1109/ECCE64574.2025.11013425
- **Investigation of varied Doping of ZnO as an ETL for $KSnI_3$ based Perovskite Solar Cell with ETL and Absorber Thickness Optimization**.
  Md Jahidur Rahman, Md Inzamam Ul Hoque, Mohammad Iftekher Ebne Jalal, Md Adnan Faisal Siddique, **Udhay Chowdhury**, Umme Mabrura Umama.
  4th International Conference on Electrical, Computer and Communication Engineering (ECCE 2025), CUET, Bangladesh.
  DOI: 10.1109/ECCE64574.2025.11013548

Honours & Achievements

- 1st Runners Up, "Discussion with the Changemaker" (CUET Trailblazers) – proposed a waterlogging solution for Chittagong City.
- 2nd Runners Up, "Face the Case 3.0" (Team Tetris) – business case on electric vehicles in Bangladesh; prize: 10,000 TK.
- IEEE R10 Ethics Champion – IEEE R10 Ethics Super Power Challenge (scored $\geq$70% in ethics assessment).
- Honourable Mention – IEEE Day Poster Presentation (Team INDUCTION MOTOR).
- Merit scholarship for exceptional performance in Secondary School Certificate examination and Higher Secondary School Certification examination.

Technical Skills

- **Simulation & Modeling:** SCAPS-1D; triboelectric nanogenerator and perovskite solar cell device modeling.
- **Programming:** Python (NumPy, Pandas, scikit-learn), C (basic)
- **Machine Learning & Data:** Random Forest, XGBoost, LightGBM; data collection, preprocessing, and model evaluation for device-performance prediction.
- **Embedded Systems & Hardware:** Microcontroller-based system design, Arduino-based sensor interfacing, thermistor and IR-based sensing prototypes, basic measurement and instrumentation.
- **Tools:** LaTeX, Git/GitHub, MS Office.

Language Skills

- Bangla – **Native**.
- English – Advanced reading, writing, and communication; IELTS/TOEFL not taken yet.
- German – Beginner; currently self-learning.

Membership

- IEEE Student Member, Bangladesh Section, Member #99931016 (Member for 2 years).